# Project 6 (Bonus) — LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline

**Title:** LangChainPro Branching Parallel And Retry Patterns For A Real Customer Support Pipeline

**Type:** Single LCEL pipeline (no agent loop, no LangGraph) · **Stack:** Python, LangChain (LCEL), `langchain-openai`, Pydantic · **Level:** Intermediate

**Tagline:** A customer-support ticket pipeline built only from LCEL pieces — a structured classifier, a reply chain per category, parallel lookups, a safe fallback and a live-streamed reply — proving plain LangChain is enough for a fixed multi-step job.

> This is the **reading guide** — the story, the folder structure and why each part exists, setup, the 4 build steps, and which document to read for each idea. The other file, README.md, is the **repo README** — you copy it into your repo folder at the end and push it to GitHub. You write every folder, file and line of code yourself; the only thing already created for you is the empty repo folder `LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline/` .

## Charter (what this project is)

A terminal pipeline that works through a queue of support tickets: classify each one, send it to the right reply chain, look up the customer's data in parallel, survive a failing model call, stream the reply, and end with one checked record per ticket. Everything is Doc05's LCEL — `prompt | model | parser` — and nothing else. No agent loop, no graph. This project proves you can apply Doc05 to a real job, and that you know where plain LangChain stops being enough.

**Jump to:** Read first · Project structure · Setup · How testing works · Step 1 · Step 2 · Step 3 · Step 4 · Run and test · Checklist · Publish

## The Story — what you're actually building

A support ticket lands in a queue: "I was charged twice for my Pro plan this month." Before any person reads it, a lot can already happen. First, decide what kind of ticket it is — billing, technical or general — so it goes to the right kind of reply. Second, for billing and technical tickets, pull up the customer's account and recent orders — both at once, since neither lookup needs the other. Third,

write the reply in a way that survives a bad minute at the API, instead of crashing the queue. Fourth, show the reply live to the support agent watching, and save one checked record the ticketing system can store — including "a human should look at this", and why.

Each of those is one Doc05 idea, applied: a structured-output chain, a plain function as a step, a `dict` of steps that runs in parallel, `.with_fallbacks()`, and `.stream()`. It's a straight line from ticket to record — no step ever goes back, waits for a person, or remembers anything after the run. That's exactly why it doesn't need LangGraph.

> **Before you read further — think about it yourself:** the fourth sample ticket reads half like billing and half like a bug. Should the classifier quietly pick one and move on, or should that doubt be visible at the end? And when the model is down and a double-charge ticket gets a canned "we got your message" reply — is that acceptable, or should it fail loudly? Decide before Step 3.

**What you're building, in one line:** a support-ticket pipeline made only of LCEL chains — classify, look up, reply, record.

**Why this needs to exist:** to prove, with working code, that a fixed multi-step job doesn't need a heavier framework just because one exists.

**When you'd reach for this at a real job:** requests with a small, fixed set of categories that never need to loop, pause or remember across restarts — support queues, alert triage, form routing.

**How it works, mechanically:** every piece is a `Runnable` joined with `|`. The category picks the reply chain; a `RunnableParallel` runs the lookups; `.with_fallbacks()` wraps each chain; `.stream()` prints the reply; a Pydantic `TicketRecord` holds the result.

**Why not plain Python and raw SDK calls?** You could — Project 1 did. But then the parallel lookups, the fallbacks and the streaming are all code you write and debug yourself, for every chain. **Why not a `try/except` around every call?** Step 3's reasoning shows what that costs versus `.with_fallbacks()`.

## Where This Fits

This is a bonus project, outside the main 5-project arc. It comes right after Doc05, because Doc05's own Build Task is one small chain module — this is where those same pieces do a real job together.

```
Project 1                → raw SDK, Python router (Docs 01-04)
Project 6 (you are here) → LangChain LCEL only, one straight pipeline
                           (Doc05)
```

```
Project 3                → LangGraph: loops, pauses, saved state
                           (Docs 08-10)
```

The line between the two frameworks, said simply: LCEL is enough while the path is fixed — even with one branch point and one parallel step. A graph earns its cost when a step must route **back**, loop until a condition is met, or pause for a person and resume later. None of that happens here. (Look ahead, optional: Doc09 — LangChain vs. LangGraph vs. RAG.)

## What to Read First (docs this project uses)

Everything this project needs was taught in Docs 01-05 and Project 1:

| You need to understand… | Read this | Used in |
|---|---|---|
| venv, `.env`, never committing secrets | Doc01 — `.env` files | Setup |
| `load_config()`, `get_logger()` | Doc01 Build Task | Step 1 |
| `prompt \| model \| parser` | Doc05 — your first LCEL chain | Steps 1, 2 |
| `.with_structured_output(Model)` | Doc05 — structured parser | Step 1 |
| Catching `ValidationError` by name | Doc05 — structured parser | Step 1 |
| Extra template keys are ignored | Doc05 — template variables | Step 1 |
| A function that returns a `Runnable` | Doc05 Build Task | All steps |
| `Literal` fields, "shape during" | Doc04 — Structured output | Step 1 |
| A router that logs its reason | Project 1 — Step 4 | Step 1 |
| Functions and dicts as steps, `RunnableParallel` | Doc05 — Real-world (part 2) | Step 2 |
| Reading a JSON file | Doc02 — JSON | Step 1 |
| Which errors are transient | Doc04 — Error types | Step 3 |
| Backoff, one retry layer only | Doc02 — Retrying | Step 3 |
| `.with_fallbacks()`, `RunnableLambda`, `.stream()` | Doc05 — Real-world (part 2) | Steps 3, 4 |
| Streaming with `flush=True` | Doc04 — Streaming | Step 4 |

Reused code, unchanged except for its folder and import lines: `config.py`, `exceptions.py`, `utils/logger.py` (Doc01) and `utils/prompts.py` (Project 1). `models/llm.py` is Doc05's Build Task rule — the model is chosen in one place — as a small `get_model()`.

Nothing here is new: Doc05's Real-world (part 2) exercise already practiced `RunnableParallel` (Step 2), `.with_fallbacks()` with `RunnableLambda` (Step 3), and `.stream()` on a chain (Step 4). Each step reuses its exercise's shape.

## Project Structure (the professional layout)

Same rule as Project 1: only folders with a real job. The new one is `chains/` — Project 1 left it out because it had no LangChain; here every piece of work is a chain.

### Names: the repo folder vs. the Python package

- **Repo folder:** `LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline/` — already created, and the exact name of the GitHub repo.
- **Python package:** `src/`, right next to `main.py`. It holds all the app's code, and every file imports from it the same way: `from src.services.pipeline import handle_ticket`.
- **Files:** `snake_case.py`, one job each; test files start with `test_`.
- **Prompt files:** named after the category they answer (`billing.txt` answers billing tickets) — the code relies on that.

### The final layout (after Step 4)

```
<repo folder>/   (LangChainPro-Branching-....-Customer-Support-Pipeline/)
├── data/
│   └── sample/
│       └── tickets.json       the 4 test tickets
├── scripts/
│   └── time_lookups.py        one after another vs. parallel
├── src/
│   ├── chains/
│   │   ├── __init__.py
│   │   ├── classifier.py      classify_ticket() -> TicketCategory
│   │   ├── fallback.py        with_canned_reply(), with_unsure_category()
│   │   ├── lookups.py         build_lookup_step() (RunnableParallel)
│   │   └── replies.py         build_reply_chain(category)
│   ├── models/
│   │   ├── __init__.py
│   │   └── llm.py             get_model()
│   ├── prompts/
│   │   ├── billing.txt        the billing reply rules
│   │   ├── classifier.txt     the sorting rules
│   │   ├── general.txt        the general reply rules
│   │   └── technical.txt      the technical reply rules
│   ├── services/
│   │   ├── __init__.py
```

```
│   │   └── pipeline.py        handle_ticket() -> TicketRecord
│   ├── utils/
│   │   ├── __init__.py
│   │   ├── logger.py          get_logger()                (Doc01)
│   │   └── prompts.py         load_prompt(name)        (Project 1)
│   ├── __init__.py
│   ├── config.py              Config, load_config()       (Doc01)
│   ├── exceptions.py          MissingConfigError          (Doc01)
│   └── schemas.py             TicketCategory, TicketRecord
├── tests/
│   ├── integration/
│   │   └── test_live_pipeline.py real API calls
│   └── unit/
│       ├── test_config.py     no API key needed
│       ├── test_fallback.py   no API key needed
│       ├── test_lookups.py    no API key needed
│       ├── test_review.py     no API key needed
│       └── test_schemas.py    no API key needed
├── .env                       your real key — never committed
├── .env.example               the setting names, no real values
├── .gitignore                 keeps .env, .venv/, __pycache__/ out
├── LICENSE                    lets others reuse your code (MIT)
├── main.py                    runs the ticket queue — the file you run
├── README.md                  the repo's front page (copied in at end)
└── requirements.txt           the libraries, for pip install -r
```

### Why each folder exists

- **`data/sample/`** — the 4 test tickets, shared by `main.py` and the integration test.
- **`scripts/`** — `time_lookups.py`, a measurement you read, not a pass/fail test.
- **`src/`** — the package, right next to `main.py`. `schemas.py` sits at its top because the chains, the service and the tests all use it.
- **`chains/`** — one file per chain job: classify, reply, look up, fall back. **Why:** each chain can be built and tested alone, and the files mirror the pipeline's steps.
- **`models/`** — builds the chat model. **Why:** every chain gets the same model; changing it is one line.
- **`prompts/`** — one `.txt` per chain. **Why:** reword how billing replies sound without touching Python.
- **`services/`** — the flow: classify → reply → record. **Why:** no `print()` of results or `input()` here beyond the live stream, so `main.py` and the tests call the exact same `handle_ticket()`.
- **`utils/`** — the logger and the prompt loader, used everywhere.
- **`tests/integration/`** — the real classifier and pipeline. **Why:** costs money, so it's run on purpose.

- **`tests/unit/`** — config, schemas, lookups, fallbacks and review rules, with no API key. **Why:** fast and free — run after every change.

### Why each root file exists

- `.env` — your real API key; never committed.
- `.env.example` — which settings to create, with no real values.
- `.gitignore` — keeps `.env`, the venv and cache files out of git.
- `LICENSE` — the MIT license text, created in Setup. Without one, nobody may legally reuse your code.
- `main.py` — the file you run: the ticket queue.
- `README.md` — what someone sees first on GitHub.
- `requirements.txt` — the library list for `pip install -r requirements.txt`.

### Why each file exists (what and why)

Every file you'll write, in one place — the same What/Why list the documents give after each tree. The steps below build these files one at a time; come back here whenever you forget why a file exists.

- `data/sample/tickets.json`
  - **What:** A JSON list of 4 test tickets, each with a `customer_id`, the `ticket` text and the `expected` category.
  - **Why:** The same inputs every run, shared by the app and the tests.
- `scripts/time_lookups.py`
  - **What:** One `main()` that times the two lookups one after the other, then with the parallel step, and prints both times.
  - **Why:** It shows you, with real numbers, how much time `RunnableParallel` saves.
- `chains/classifier.py`
  - **What:** `build_classifier_chain()` returns `prompt | get_model().with_structured_output(TicketCategory)`, wrapped (from Step 3) with `with_unsure_category()`; `classify_ticket(ticket)` invokes it and returns the `TicketCategory`.
  - **Why:** Each kind of ticket needs a different reply. Sorting first means each reply prompt stays short and focused.
- `chains/fallback.py`
  - **What:** `TRANSIENT_ERRORS` (connection errors and timeouts, rate limits, server errors), a `FALLBACK_REPLY` text, `canned_reply()` and `unsure_category()` fallback functions, and `with_canned_reply(chain)` / `with_unsure_category(chain)`, which call `chain.with_fallbacks([...], exceptions_to_handle=TRANSIENT_ERRORS)`.

    - **Why:** During an outage the customer still gets a polite reply, while a bad key or a code bug still fails loudly.
- `chains/lookups.py`
    - **What:** Fake `FAKE_ACCOUNTS` and `FAKE_ORDERS` data, three small functions that read the inputs (`get_ticket_text`, `get_account_info`, `get_order_history`), and `build_lookup_step()`, which returns a `RunnableParallel` running the lookups at the same time.
    - **Why:** Two lookups that don't depend on each other finish faster together than one after the other.
- `chains/replies.py`
    - **What:** `build_reply_chain(category)`: load the prompt file with the category's name, build `lookups | prompt | model | parser` (general has no lookups), and wrap it with `with_canned_reply()`.
    - **Why:** Each category gets its own rules, so billing and technical replies each stay on topic.
- `models/llm.py`
    - **What:** A `MODEL_NAME` constant and one function, `get_model()`, which returns `ChatOpenAI(model=MODEL_NAME, temperature=0)`.
    - **Why:** Every chain gets the same model and settings from here. To try another model you change one line, not every file.
- `prompts/*.txt`
    - **What:** Four text files: the classifier's sorting rules, and one reply prompt per category (billing, technical, general).
    - **Why:** You can change the rules for one kind of ticket without touching the code or the other prompts.
- `services/pipeline.py`
    - **What:** `stream_reply(chain, inputs)` prints each chunk from `chain.stream(inputs)` and returns the full text; `review_reasons(category, used_fallback)` returns the list of reasons for a human check; `handle_ticket(ticket, customer_id)` classifies, builds the reply chain, streams the reply, and returns a `TicketRecord`.
    - **Why:** The whole flow is one function that `main.py` and the tests can call, and the reply appears word by word instead of all at once.
- `utils/logger.py`
    - **What:** One function, `get_logger(name)`. It gets the logger for that name and, only the first time (`if not logger.handlers:`), adds a console handler with a timestamp format and the level from `LOG_LEVEL`.
    - **Why:** Every file logs in the same format with one line of code. Without the `if not logger.handlers:` guard, every log line would print twice.
- `utils/prompts.py`
    - **What:** One constant, `PROMPTS_DIR` (`src/prompts`), and one function, `load_prompt(name)`, which opens `PROMPTS_DIR + "/" + name + ".txt"` and returns

its text.
    - **Why:** Prompts live in text files, so you can change how a chain behaves without editing Python.
- `__init__.py` (every code folder)
    - **What:** An empty file. You create one in every folder that holds Python code — nothing goes inside it.
    - **Why:** Python only treats a folder as a package when this file is there. Without it, an import like `from src.<folder> import ...` fails with `ModuleNotFoundError`.
- `config.py`
    - **What:** A `Config` dataclass with two fields (`openai_api_key`, `log_level`), a helper `require_env(key)` that raises `MissingConfigError` when a setting is empty, and `load_config()`, which calls `load_dotenv()`, reads both settings and returns a `Config`.
    - **Why:** Every file gets its settings from one place. If the key is missing, the app stops at startup with one clear line.
- `exceptions.py`
    - **What:** A few small classes that inherit from `Exception` — at least `MissingConfigError`. Each one is just `class Name(Exception): pass`, with a docstring saying when it's raised.
    - **Why:** Code can catch one specific problem (`except MissingConfigError:`). Without named errors you'd catch every `Exception`, and real bugs would be hidden along with the expected problem.
- `schemas.py`
    - **What:** Two Pydantic classes: `TicketCategory` (`category` as `Literal["billing", "technical", "general"]`, `reason`, `sure: bool`) and `TicketRecord` (customer, ticket, category, reply, `used_fallback`, `needs_human_review`, `review_reasons`). The three fields added after Step 1 have defaults, so older code still builds a record.
    - **Why:** The classifier's answer is checked, and every ticket's result has the same fields — easy to print, store or review.
- `main.py`
    - **What:** `main()` loads the config, reads `tickets.json`, runs `handle_ticket()` for every ticket, and prints each `TicketRecord` with `print_record()`.
    - **Why:** One starting point that runs the ticket queue and prints the results. The chains live in other files.
- `tests/integration/test_live_pipeline.py`
    - **What:** Test functions against the real API, each printing `OK` or `WRONG`: each of billing/technical/general tickets gets its expected category, a billing ticket runs all the way to a reply, and the unclear ticket is flagged for a human.
    - **Why:** It proves each ticket gets the right category and a real reply.
- `tests/unit/test_config.py`
    - **What:** Three test functions, no API call, each printing `OK` or `WRONG`: `require_env()` raises for a missing and an empty setting (a made-up name, so your `.env` is untouched),

and `load_config()` returns a `Config`.
    - **Why:** It proves a missing key stops the app with one clear error, with no API call.
- `tests/unit/test_fallback.py`
    - **What:** A `ChatOpenAI` pointed at a port where nothing listens, plus a working step and a buggy step. Three cases, each `OK` / `WRONG`: a real outage gives `FALLBACK_REPLY`, a working chain never uses it, and a `KeyError` is still raised.
    - **Why:** It proves the fallback fires during an outage and never hides a bug — with no key and no cost.
- `tests/unit/test_lookups.py`
    - **What:** Three cases, each `OK` / `WRONG`: a known customer gets account and order data, an unknown customer gets the "not found" texts, and `build_lookup_step()` returns the keys `ticket`, `account`, `orders` in under 1.5 seconds.
    - **Why:** It catches a change that quietly makes the lookups run one after another again.
- `tests/unit/test_review.py`
    - **What:** A helper `make_category(sure)` and three cases, each `OK` / `WRONG`, calling `review_reasons()`: a sure ticket gets no reasons, an unsure ticket gets one, and a fallback reply gets one.
    - **Why:** Each review rule is checked alone, so a rule that stops firing is noticed at once.
- `tests/unit/test_schemas.py`
    - **What:** Three cases, each `OK` / `WRONG`: a good `TicketCategory` builds, `shipping` raises `ValidationError`, and a `TicketRecord` builds from the 4 Step 1 fields.
    - **Why:** It proves the classifier can never return a category no branch handles, and that old records still build after new fields are added.

### How imports work in this layout

- Every folder with Python files gets an empty `__init__.py` (not `prompts/` or `data/`). It marks the folder as a package, so `src.chains.replies` is importable.
- Python looks for imports in the folder you **run from**. `main.py` and `src/` sit side by side in the repo folder, so `python main.py` finds `from src.services.pipeline import handle_ticket` right there.
- Tests and scripts live in sub-folders, so run them as modules from the repo folder: `python -m tests.unit.test_lookups`, `python -m scripts.time_lookups`.
- Always run commands from the **repo root** — the prompts folder and `data/sample/tickets.json` are found from there.

### Left out on purpose (and when they come)

| Folder / file in a big template | Why not here | Where it appears |
|---|---|---|
| `graph.py`, `state.py` (LangGraph) | No loops, no pauses, no saved state | Project 3 (Doc09) |
| `RunnableBranch` | An `if` and a prompt file per category are clearer | Only if branches grow into whole different chains |
| `.with_retry()` | The OpenAI client already retries; a second layer stacks | If you ever turn the client's retries off |
| `.batch()` | One ticket at a time is fine for a demo queue | When the queue gets large |
| `api/`, a real database | A terminal app with stand-in lookups | Project 5 (Doc12) |
| `Dockerfile`, CI | Nothing to deploy yet | Doc12-13 |
| pytest, `conftest.py` | Plain test scripts, same as Project 1 | Doc13 |
| A frontend | Streaming works in the terminal | Streamlit, when a project needs one |

### How the structure grows, step by step

| Step | New files | Changed files |
|---|---|---|
| Setup | root files, `src/__init__.py` | — |
| 1 | `config.py`, `exceptions.py`, `utils/logger.py`, `utils/prompts.py`, `models/llm.py`, `schemas.py`, `prompts/` (4 files), `chains/classifier.py`, `chains/replies.py`, `services/pipeline.py`, `main.py`, `data/sample/tickets.json`, `tests/unit/test_config.py`, `tests/unit/test_schemas.py`, `tests/integration/test_live_pipeline.py` | — |
| 2 | `chains/lookups.py`, `scripts/time_lookups.py`, `tests/unit/test_lookups.py` | `prompts/billing.txt`, `prompts/technical.txt`, `chains/replies.py` |
| 3 | `chains/fallback.py`, `tests/unit/test_fallback.py` | `schemas.py`, `chains/classifier.py`, `chains/replies.py`, `services/pipeline.py`, `main.py` |
| 4 | `tests/unit/test_review.py` | `schemas.py`, `services/pipeline.py`, |

| Step | New files | Changed files |
|---|---|---|
| | | `main.py`, `test_live_pipeline.py` |

## Setup (do this once, before Step 1)

**Read first:** Doc01 — Virtual environments, Doc01 — `.env` files. The same steps as Project 1, with a different library list.

1. Go into the repo folder, start git, and make a venv. From the curriculum's root folder ( `multi-agent-ai-engineering-curriculum/` ):

```
cd project_6_langchainpro_lcel_patterns
# the \ joins the two lines into one folder name
cd LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-\
Customer-Support-Pipeline
git init
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
```

2. Create `.gitignore` **before** your first commit:

```
# secrets — never pushed
.env
# the virtual environment — rebuilt from requirements.txt
.venv/
# Python's compiled cache files
__pycache__/
*.pyc
```

3. Create `.env` and `.env.example`, the same as Project 1:

```
# .env — never commit this file
OPENAI_API_KEY=sk-your-real-key-here
LOG_LEVEL=INFO
```

```
# .env.example — commit this one
# Copy this file to .env and fill in the real values. Never commit .env.
OPENAI_API_KEY=
LOG_LEVEL=INFO
```

4. Create `requirements.txt`:

```
langchain-core>=1.0
langchain-openai>=1.0
pydantic>=2.0
python-dotenv>=1.0
```

New since Project 1: `langchain-core` (prompts, parsers, `RunnableParallel`) and `langchain-openai` (`ChatOpenAI`, which brings `openai` with it) — Doc05's packages topic.

5. Create an empty `src/__init__.py`, then install:

```
pip install -r requirements.txt
```

**Check it worked:** from the repo folder, `python -c "import src; print('ok')"` prints `ok`. It works because the `src/` folder sits right there — that's why every command runs from the repo folder.

6. Create `LICENSE` — the MIT license, so others may reuse your code. Put the current year and your own name on the Copyright line:

```
MIT License

Copyright (c) 2026 Your Name

Permission is hereby granted, free of charge, to any person obtaining a copy
of this software and associated documentation files (the "Software"), to deal
in the Software without restriction, including without limitation the rights
to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
copies of the Software, and to permit persons to whom the Software is
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IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.
```

## A Real Example (so this isn't just theory)

**Scenario:** SupportCo is a small SaaS company. Step 1 saves these 4 tickets in `data/sample/tickets.json`:

1. `C-1001`: "I was charged twice for my Pro plan this month (INV-2201)." → **billing**.
2. `C-1002`: "The export button spins forever and never downloads." → **technical**.
3. `C-1003`: "Do you have a referral program for other companies?" → **general** (no lookups needed).
4. `C-1004`: "My invoice and the dashboard numbers don't match. Bug?" → **billing or technical** — the honest answer is "not sure", and the pipeline must show that.

**The lookups** (Step 2) are stand-ins for a real database: `get_account_info()` returns a plan and a signup date, `get_order_history()` returns recent invoices. Each waits 1 second on purpose, like a real network call.

## Why This Project Is Good for Your Portfolio

- **What problem it solves:** a real multi-step pipeline — routing, concurrency, failure handling, streaming and a checked record — in plain LangChain.
- **Why it matters:** knowing when you *don't* need a heavier framework is as valuable as knowing the framework.
- **When you'd build something like this at a job:** any ticket, alert or form pipeline with a few fixed categories.
- **How it's built:** a structured classifier, a reply chain per category, `RunnableParallel` lookups, `.with_fallbacks()` on transient errors only, `.stream()`, and a Pydantic record with review reasons.

**Problems you'll likely run into, and how to fix them:**

| Problem | Fix |
|---|---|
| The unclear ticket silently becomes "billing" and nobody notices | Keep the classifier's `sure` flag and turn it into a review reason (Step 4) |
| The parallel lookups take as long as sequential ones | Check the two functions don't call each other, and measure with `time_lookups.py` |
| `KeyError: 'account'` in the billing prompt | The step before the prompt must return a dict with exactly the prompt's keys |
| The fallback hides a bad key or a bug | Pass `exceptions_to_handle=` — only transient errors fall back |

| Problem | Fix |
| --- | --- |
| A failed call takes far longer than expected | Two retry layers are stacked — keep only the client's retries |
| The streamed reply and the record disagree | Build the record from the same joined reply, never from a second call |
| `ModuleNotFoundError: No module named 'src'` | The venv isn't active, or you ran from the wrong folder — run from the repo root |

## Built During These Documents

05_langchain_fundamentals (on top of Docs 01-04 and Project 1)

## Plan Before You Code

See 15_five_projects_index: write down Problem → Requirements → Architecture → Components → Data Flow → Implementation Plan → Coding Tasks. Before coding, write one sentence for what each step proves — if you can't, you'll build all four patterns at once and never know which one does the work.

## How To Build This — Step by Step

The project grows step by step, and **after every step it runs**: Step 1 classifies and replies, Step 2 adds parallel lookups, Step 3 survives a failing model, Step 4 streams and records. Build it yourself first; use the hints if you're stuck, and the solution only to compare.

Every command runs from the **repo root** with the venv active.

### How testing works in this project

**Story:** LangChainPro's promise is "the right reply reaches the right ticket, fast, and survives a bad minute at the API" — a promise a casual run can't confirm. So every step adds the tests for the part it builds: the classifier, the parallel lookups, the fallback, the review rules. When something breaks, the newest `WRONG` points straight at the newest code. **If not:** a prompt change could quietly misroute tickets, or the fallback could hide a real bug, and you'd only find out from a customer.

- **Two kinds of test:**
  - `tests/unit/` — code that needs no model: the settings, the schemas, the lookups, the fallback, the review rules. No API key, no cost, under a second each (the lookup test waits ~1 s on purpose). **When:** after every change to the file it tests.

    - `tests/integration/test_live_pipeline.py` — the real classifier and pipeline against the API, one case per ticket kind. Real API calls. **When:** at the end of a step, and after changing a prompt or the classifier.
- **One pattern for every case** — so the output never needs judging:

```
def test_something() -> None:
    result = "OK"
    if <it came out wrong>:
        result = "WRONG"
    print(f"case 1 (what it checks) -> {result}")
```

- **An error you expect** (like `MissingConfigError` or a `ValidationError`): start with `result = "WRONG"`, and set `result = "OK"` inside `except TheError:`.
- **One function per case**, all called at the bottom under `if __name__ == "__main__":`. Each file numbers its own cases, and case numbers never change once written.
- **Plain scripts, not pytest** — this project uses only Doc05, which teaches plain `print()` tests (the same style as Project 1). Run each with `python -m`.
- **Run from the repo root with `-m`:** `python -m tests.unit.test_lookups` (dots, no `.py`). Running the file by its path fails with `No module named 'src'`.
- **Got a `WRONG`?** Read that one test function, then the code it calls. Fix the code — change the test only if the test itself checks the wrong thing.

### Step 1 — A Classifier That Sends Each Ticket to the Right Reply Chain

*Project: LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline — Step 1 of 4*

- **Where you are now:** Setup is done — nothing runs yet.
- **What's new in this step:** the shared files (config, errors, logger, prompt loader, model), `TicketCategory` and `TicketRecord`, a classifier chain, a reply chain per category, `handle_ticket()`, the sample tickets, `main.py`, and the first tests (`test_config.py`, `test_schemas.py`, `test_live_pipeline.py`).
- **What stays the same:** — (first step).
- **Why:** every later step makes one part stronger. Prove the basic shape first — one chain decides, a second chain, picked by that decision, replies.
- **How:** `classify_ticket()` (`prompt | model.with_structured_output(TicketCategory)`) → `build_reply_chain(category)` (the category names the prompt file) → `.invoke()` → a `TicketRecord` → `print_record()`.
- **Working after this step:** `python main.py` prints a category and a reply for each of the 4 tickets, and the test files print `OK`.

**Read first:** Doc05 — your first LCEL chain, Doc05 — structured parser, Doc05 — parser failure, Doc01 Build Task.

**Stuck on this step?** Hint 1 · Hint 2 · Show me the solution

What to do:

1. Bring in the shared files: `config.py`, `exceptions.py`, `utils/logger.py`, `utils/prompts.py` (folder `src/prompts`), and `models/llm.py` with a no-argument `get_model()`.
2. Write `schemas.py`: `TicketCategory` (`category` as a `Literal` of the 3 names, `reason`, `sure`) and `TicketRecord` (`customer_id`, `ticket`, `category`, `reply`).
3. Write `prompts/classifier.txt` and one reply prompt per category: `billing.txt`, `technical.txt`, `general.txt`.
4. Write `chains/classifier.py` (`build_classifier_chain()`, `classify_ticket()` — catch `ValidationError`, fall back to "general, not sure", log the decision) and `chains/replies.py` (`build_reply_chain(category)`).
5. Write `services/pipeline.py` (`handle_ticket()` prints the reply and returns a `TicketRecord`), `data/sample/tickets.json`, and `main.py` (`print_record()`; loop over the samples; catch `AuthenticationError` and `RateLimitError`).
6. Write `tests/unit/test_config.py`, `tests/unit/test_schemas.py` and `tests/integration/test_live_pipeline.py`. Run everything, and look closely at ticket 4.

The root files — `.gitignore`, `.env`, `.env.example`, `requirements.txt` and `LICENSE` — come from Setup; this step adds the files below.

**Your files after Step 1:**

```
<repo folder>/
├── data/
│   └── sample/
│       └── tickets.json    (new) the 4 test tickets
├── src/
│   ├── chains/
│   │   ├── __init__.py     (new)
│   │   ├── classifier.py   (new) classify_ticket()
│   │   └── replies.py      (new) build_reply_chain(category)
│   ├── models/
│   │   ├── __init__.py     (new)
│   │   └── llm.py          (new) get_model()
│   ├── prompts/
│   │   ├── billing.txt     (new)
│   │   ├── classifier.txt  (new)
│   │   ├── general.txt     (new)
│   │   └── technical.txt   (new)
```

```
│   ├── services/
│   │   ├── __init__.py     (new)
│   │   └── pipeline.py     (new) handle_ticket() -> TicketRecord
│   ├── utils/
│   │   ├── __init__.py     (new)
│   │   ├── logger.py       (new)
│   │   └── prompts.py      (new)
│   ├── __init__.py
│   ├── config.py           (new)
│   ├── exceptions.py       (new)
│   └── schemas.py          (new) TicketCategory, TicketRecord
├── tests/
│   ├── integration/
│   │   └── test_live_pipeline.py (new) real API calls
│   └── unit/
│       ├── test_config.py  (new) no API key needed
│       └── test_schemas.py (new) no API key needed
└── main.py                 (new) runs the 4 sample tickets
```

**Why these files (what and why):**

- `chains/classifier.py`
  - **What:** `build_classifier_chain()` (`prompt | structured model`) and `classify_ticket()`, which runs it, catches `ValidationError`, and logs the decision.
  - **Why:** Each kind of ticket needs a different reply; sorting first keeps each reply prompt focused.
- `chains/replies.py`
  - **What:** `build_reply_chain(category)` — loads the prompt file named after the category and returns `prompt | model | parser`.
  - **Why:** Each category gets its own rules without three near-identical functions.
- `models/llm.py`
  - **What:** `MODEL_NAME` and `get_model()` → `ChatOpenAI(model=MODEL_NAME, temperature=0)`.
  - **Why:** Every chain gets the same model; changing it is one line.
- `services/pipeline.py`
  - **What:** `handle_ticket(ticket, customer_id)` — classify, pick the reply chain, reply, return a `TicketRecord`.
  - **Why:** One function `main.py` and the tests both call.
- `prompts/*.txt`
  - **What:** the classifier's rules and one reply prompt per category.
  - **Why:** reword a kind of ticket without touching code.
- `schemas.py`

    - **What:** `TicketCategory` (3-value `Literal`) and `TicketRecord` (the 4 Step 1 fields; later fields have defaults).
    - **Why:** the classifier's answer is checked, and every result has the same fields.
- `config.py`, `exceptions.py`, `utils/logger.py`, `utils/prompts.py`
    - **What:** Doc01/Project 1's config, named errors, logger and prompt loader.
    - **Why:** settings fail loudly at startup, logging is uniform, prompts live in files.
- `main.py`
    - **What:** loads the config, reads the tickets, runs `handle_ticket()` on each, prints each record.
    - **Why:** one starting point for the queue.
- `data/sample/tickets.json`
    - **What:** the 4 test tickets, including the unclear one.
    - **Why:** the same inputs every run, shared with the tests.
- `tests/unit/test_config.py`, `test_schemas.py`; `tests/integration/test_live_pipeline.py`
    - **What:** the first tests — settings, schemas (no key) and the real classifier/pipeline.
    - **Why:** prove the foundation before anything builds on it.

**Tests for this step — write them yourself** (the solution has the code, to compare):

- **cases 1-3 (settings)** — `tests/unit/test_config.py`
    - **Check:** `require_env()` raises `MissingConfigError` for a missing and an empty made-up setting, and `load_config()` returns a `Config`.
    - **Why:** a missing key must stop the app at startup with one clear line.
    - **When:** after any change to `config.py` or `exceptions.py`.
    - **If not:** a missing key would surface later as a confusing `AuthenticationError`.
- **case 1 (good category)** — `tests/unit/test_schemas.py`
    - **Check:** a `TicketCategory` with `billing` builds.
    - **Why:** the happy path must work before the error path means anything.
    - **When:** after changing `schemas.py`.
    - **If not:** a broken schema would fail every ticket silently.
- **case 2 (unknown rejected)** — `tests/unit/test_schemas.py`
    - **Check:** `category="shipping"` raises `ValidationError`.
    - **Why:** the classifier must never return a category no branch handles.
    - **When:** after changing the `Literal`.
    - **If not:** a typo category would route to nothing.
- **case 3 (record builds)** — `tests/unit/test_schemas.py`
    - **Check:** a `TicketRecord` builds from the 4 Step 1 fields.
    - **Why:** later steps add fields with defaults — this proves old records still build.

    - **When:** after adding a field to `TicketRecord`.
    - **If not:** a new required field would break every earlier caller.
- **cases 1-4 (classification + pipeline)** — `tests/integration/test_live_pipeline.py`
    - **Check:** billing/technical/general tickets get their expected category, and a billing ticket runs to a non-empty reply.
    - **Why:** proves the key, the model, the prompts and the wiring work together.
    - **When:** at the end of the step, and after changing a prompt or the classifier.
    - **If not:** a misrouting prompt would be found by a customer, not by you.

### Step 2 — Customer Lookups That Run at the Same Time

*Project: LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline — Step 2 of 4*

- **Where you are now:** after Step 1 every ticket gets a category and a reply — but billing and technical replies know nothing about the customer.
- **What's new in this step:** `chains/lookups.py` (two fake, 1-second lookups and a `RunnableParallel` step), `{account}` and `{orders}` in two prompts, a timing script and a test.
- **What stays the same:** the classifier, `handle_ticket()`, `main.py`, and the general reply chain.
- **Why:** the two lookups don't need each other. Running them one after another wastes a full second per ticket, for nothing.
- **How:** `RunnableParallel({"ticket": ..., "account": ..., "orders": ...})` — every function gets the same input, they run at the same time, and the output dict fills the prompt. It goes in front of the billing and technical chains with `|`.
- **Working after this step:** replies use the account data, and `python -m scripts.time_lookups` shows about 2.0 s vs. 1.0 s.

**Read first:** Doc05 — Real-world (part 2) — `RunnableParallel`, a plain function as a step, a `dict` that runs its values in parallel.

**Stuck on this step?** Hint 1 · Hint 2 · Show me the solution

What to do:

1. Add `{account}` and `{orders}` to `prompts/billing.txt` and `prompts/technical.txt`.
2. Write `chains/lookups.py`: `get_ticket_text()`, `get_account_info()`, `get_order_history()` (fake data, `time.sleep(1)` each) and `build_lookup_step()`.
3. Change `build_reply_chain()`: billing and technical start with `build_lookup_step() |`; general stays as it was.
4. Write `scripts/time_lookups.py` and `tests/unit/test_lookups.py`, and run both.

**Your files after Step 2:**

```
<repo folder>/
├── scripts/
│   └── time_lookups.py     (new) one after another vs. parallel
├── src/
│   ├── chains/
│   │   ├── lookups.py      (new) the lookups + build_lookup_step()
│   │   └── replies.py      (changed) lookups before billing/technical
│   └── prompts/
│       ├── billing.txt     (changed) + {account}, {orders}
│       └── technical.txt   (changed) + {account}, {orders}
└── tests/
    └── unit/
        └── test_lookups.py (new) no API key needed
```

**Why these files (what and why):**

- `chains/lookups.py`
  - **What:** fake account/order data, three input-reading functions, and `build_lookup_step()` → a `RunnableParallel`.
  - **Why:** two independent lookups finish faster together than one after the other.
- `chains/replies.py`
  - **What:** billing and technical chains now start with the lookup step; general is unchanged.
  - **Why:** the billing/technical prompts need the account data; general doesn't.
- `prompts/billing.txt`, `prompts/technical.txt`
  - **What:** now have `{account}` and `{orders}` slots.
  - **Why:** the reply can use the customer's real data.
- `scripts/time_lookups.py`
  - **What:** times the lookups one-by-one vs. parallel and prints both.
  - **Why:** turns "it's faster" into a number.
- `tests/unit/test_lookups.py`
  - **What:** known/unknown customers and a parallel-timing check, all `OK` / `WRONG`.
  - **Why:** catches a change that makes the lookups serial again.

**Tests for this step — write them yourself:**

- **case 1 (known customer)** — `tests/unit/test_lookups.py`
  - **Check:** `get_account_info` and `get_order_history` return the known customer's data.
  - **Why:** the reply is only as good as the lookup.
  - **When:** after changing `lookups.py`.
  - **If not:** a wrong key would hand the reply the wrong customer's data.

- **case 2 (unknown customer)** — `tests/unit/test_lookups.py`
  - **Check:** an unknown id gets the "not found" texts.
  - **Why:** a missing customer must be handled, not crash.
  - **When:** same as case 1.
  - **If not:** an unknown id would raise a `KeyError` mid-reply.
- **case 3 (parallel keys and speed)** — `tests/unit/test_lookups.py`
  - **Check:** `build_lookup_step().invoke(...)` returns the keys `account`, `orders`, `ticket` in under 1.5 s.
  - **Why:** two 1-second lookups in ~1 s proves they ran at the same time.
  - **When:** after changing `build_lookup_step()`.
  - **If not:** the lookups could silently go back to running one after another.

### Step 3 — Retries, Then a Safe Fallback, When a Model Call Fails

*Project: LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline — Step 3 of 4*

- **Where you are now:** after Step 2 the pipeline works on a good day — but one outage, even after the client's retries, crashes the ticket.
- **What's new in this step:** `chains/fallback.py` with a canned reply and an "unsure general" category, both used through `.with_fallbacks()` — only for transient errors — and a test that forces a real outage.
- **What stays the same:** the lookups, the prompts, the reply flow.
- **Why:** real API calls fail at any real volume. The retry part already exists — the OpenAI client retries 2 times with backoff (Doc04) — so adding `.with_retry()` would stack a second layer. What's missing is what happens after the retries run out.
- **How:** `chain.with_fallbacks([RunnableLambda(canned_reply)], exceptions_to_handle=TRANSIENT_ERRORS)` around each reply chain and the classifier. `TicketRecord` gets a `used_fallback` field with a default.
- **Working after this step:** when the model can't be reached, the ticket still gets a safe reply; a bad key or a bug still fails loudly.

**Read first:** Doc04 — Error types, Doc02 — Retrying, Doc05 — Real-world (part 2) — `.with_fallbacks()`, `RunnableLambda`, "one retry layer only".

**Stuck on this step?** Hint 1 · Hint 2 · Show me the solution

What to do:

1. Write `chains/fallback.py`: `TRANSIENT_ERRORS`, `FALLBACK_REPLY`, `canned_reply()`, `unsure_category()`, `with_canned_reply(chain)`, `with_unsure_category(chain)`.
2. Wrap the classifier chain and every reply chain.

3. Add `used_fallback: bool = False` to `TicketRecord`. In `handle_ticket()`, set it (the reply equals `FALLBACK_REPLY`). In `main.py`, drop the `RateLimitError` catch and add one `print_record()` line for `used_fallback`.
4. Write `tests/unit/test_fallback.py`: a real outage falls back, a working chain doesn't, a bug still raises.

**Your files after Step 3:**

```
<repo folder>/
├── src/
│   ├── chains/
│   │   ├── classifier.py   (changed) + with_unsure_category()
│   │   ├── fallback.py     (new) backups for transient errors only
│   │   └── replies.py      (changed) + with_canned_reply()
│   ├── services/
│   │   └── pipeline.py     (changed) + used_fallback
│   ├── schemas.py          (changed) + used_fallback
│   └── (the other files, unchanged)
├── main.py                 (changed) prints used fallback
└── tests/
    └── unit/
        └── test_fallback.py (new) no API key needed
```

**Why these files (what and why):**

- `chains/fallback.py`
  - **What:** `TRANSIENT_ERRORS`, `FALLBACK_REPLY`, and `with_canned_reply` / `with_unsure_category` wrappers.
  - **Why:** an outage still gets a polite reply; a bug still fails loudly.
- `chains/classifier.py`
  - **What:** the chain is wrapped with `with_unsure_category()`.
  - **Why:** a classifier outage falls back to "general, not sure" instead of stopping the ticket.
- `chains/replies.py`
  - **What:** every reply chain is wrapped with `with_canned_reply()`.
  - **Why:** a transient failure anywhere in the chain sends the canned reply.
- `services/pipeline.py`
  - **What:** sets `used_fallback` by comparing the reply to `FALLBACK_REPLY`.
  - **Why:** a fallback reply must be visible in the record.
- `schemas.py`
  - **What:** `TicketRecord` gains `used_fallback: bool = False`.
  - **Why:** the record says whether the safe reply was sent; the default keeps earlier code working.

- `main.py`
  - **What:** drops the `RateLimitError` catch and prints `used fallback`.
  - **Why:** the fallback now handles rate limits inside the pipeline.
- `tests/unit/test_fallback.py`
  - **What:** a forced outage, a working chain and a buggy step, all `OK` / `WRONG`.
  - **Why:** proves the fallback fires on an outage and never hides a bug.

**Tests for this step — write them yourself:**

- **case 1 (outage falls back)** — `tests/unit/test_fallback.py`
  - **Check:** a chain whose model points at a dead port returns `FALLBACK_REPLY`.
  - **Why:** the whole point of the fallback is surviving a real outage.
  - **When:** after changing `fallback.py`.
  - **If not:** you'd discover the fallback is broken during a real outage.
- **case 2 (working, no fallback)** — `tests/unit/test_fallback.py`
  - **Check:** a working chain returns its real reply, not the canned one.
  - **Why:** the fallback must not fire when nothing is wrong.
  - **When:** same as case 1.
  - **If not:** every reply could silently become the canned text.
- **case 3 (bug still raises)** — `tests/unit/test_fallback.py`
  - **Check:** a `KeyError` from a buggy step is **not** caught by the fallback.
  - **Why:** `exceptions_to_handle` must cover only transient errors.
  - **When:** after changing `TRANSIENT_ERRORS`.
  - **If not:** a real bug would hide behind a polite canned reply.

### Step 4 — Stream the Reply Live and Build One Structured Record per Ticket

*Project: LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline — Step 4 of 4*

- **Where you are now:** after Step 3 the pipeline survives failures — but the reply appears all at once, and the unclear ticket and every fallback pass without a flag.
- **What's new in this step:** `stream_reply()`, `review_reasons()`, two new `TicketRecord` fields (`needs_human_review`, `review_reasons`), two new `print_record()` lines, and `tests/unit/test_review.py`.
- **What stays the same:** the classifier, the lookups, the fallbacks, and the names and return types of `handle_ticket()` and `print_record()`.
- **Why:** one reply is needed in two shapes — live text for the person watching, a checked record for the ticketing system — and both must come from the same reply.
- **How:** `chain.stream(inputs)` → print each piece with `flush=True` and join them → `review_reasons()` → the two new fields in `TicketRecord(...)`.

- **Working after this step:** the full project — each reply streams live, and each ticket ends with a record that says whether a human should review it, and why.

**Read first:** Doc05 — Real-world (part 2) ( `.stream()` on a chain), Doc04 — Streaming, Doc04 — Structured output.

**Stuck on this step?** Hint 1 · Hint 2 · Show me the solution

What to do:

1. Add `needs_human_review: bool = False` and `review_reasons: list[str] = []` to `TicketRecord`.
2. In `services/pipeline.py`, write `stream_reply(chain, inputs)` and `review_reasons(category, used_fallback)`. In `handle_ticket()`, swap `.invoke()` + print for `stream_reply()`, and fill the two new fields.
3. In `print_record()`, print the review flag and each reason.
4. Write `tests/unit/test_review.py` and add a flag check to the integration test. `test_schemas.py` stays unchanged. Run everything — ticket 4 should be flagged.

**Your files after Step 4 (final):** the full tree is in Project Structure. What changed:

```
<repo folder>/
├── src/
│   ├── services/
│   │   └── pipeline.py     (changed) + stream_reply(), review_reasons()
│   └── schemas.py          (changed) + needs_human_review, review_reasons
├── main.py                 (changed) print_record() + review lines
└── tests/
    ├── integration/
    │   └── test_live_pipeline.py (changed) + review flag
    └── unit/
        └── test_review.py  (new) no API key needed
```

**Why these files (what and why):**

- `services/pipeline.py`
  - **What:** adds `stream_reply()` (prints the reply live, returns the full text) and `review_reasons()`; `handle_ticket()` now streams and fills the two new fields.
  - **Why:** one reply serves the live view and the record, from the same text.
- `schemas.py`
  - **What:** `TicketRecord` gains `needs_human_review` and `review_reasons`, both with defaults.
  - **Why:** the record says whether — and why — a human should look.
- `main.py`

    - **What:** `print_record()` prints the review flag and each reason.
    - **Why:** a flagged ticket must look different from a clean one.
- `tests/unit/test_review.py`
    - **What:** checks each review rule with hand-made inputs, `OK` / `WRONG`.
    - **Why:** a rule that stops firing is caught at once.
- `tests/integration/test_live_pipeline.py`
    - **What:** adds a case that the unclear ticket is flagged for a human.
    - **Why:** proves the ambiguity Step 1 revealed is acted on.

**Tests for this step — write them yourself:**

- **case 1 (sure, no fallback)** — `tests/unit/test_review.py`
    - **Check:** a sure ticket with no fallback gets no review reasons.
    - **Why:** a clean ticket must not be flagged.
    - **When:** after changing `review_reasons()`.
    - **If not:** every ticket would go to a human, and the flag would mean nothing.
- **case 2 (unsure flagged)** — `tests/unit/test_review.py`
    - **Check:** an unsure ticket gets exactly one reason.
    - **Why:** the ambiguity must become a visible flag.
    - **When:** same as case 1.
    - **If not:** a guessed category would ship with no warning.
- **case 3 (fallback flagged)** — `tests/unit/test_review.py`
    - **Check:** a fallback reply gets exactly one reason.
    - **Why:** a canned reply should always get a human's eyes.
    - **When:** same as case 1.
    - **If not:** customers would get canned replies nobody reviews.
- **case 5 (unclear flagged)** — `tests/integration/test_live_pipeline.py`
    - **Check:** the unclear ticket ( `C-1004` ) ends with `needs_human_review` true.
    - **Why:** proves the whole pipeline acts on the doubt, end to end.
    - **When:** at the end of the step, and after changing the classifier or the review rules.
    - **If not:** the ambiguity would disappear silently in a real run.

**Final Deliverable: LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline** — a support-ticket pipeline in a proper package layout, built only from LCEL: a structured classifier, a reply chain per category, parallel lookups, fallbacks on transient errors only, a live-streamed reply, and a checked record per ticket.

**Why this is not an agent (said plainly):** nothing here decides what to do next at run time, and nothing loops. The path is fixed: classify, reply, record. That's the point — it's the LangChain side of the line that Project 3 crosses.

## Run, Test and Configure the Final Project

The same commands and settings as the repo's README.md — once Step 4 is done, this is how the finished project is run, tested and configured.

### Configuration

| Variable | Required | Default | Meaning |
|---|---|---|---|
| `OPENAI_API_KEY` | yes | — | Your OpenAI API key |
| `LOG_LEVEL` | no | `INFO` | `DEBUG`, `INFO`, `WARNING` or `ERROR` |

Other settings live in code, each in one place:

| Setting | Where | Default |
|---|---|---|
| Chat model | `src/models/llm.py` → `MODEL_NAME` | `gpt-4o-mini` |
| Errors that fall back | `src/chains/fallback.py` → `TRANSIENT_ERRORS` | connection, rate limit, 5xx |
| The canned reply | `src/chains/fallback.py` → `FALLBACK_REPLY` | "a member of our team will reply…" |
| Fake lookup delay | `src/chains/lookups.py` → `LOOKUP_DELAY_SECONDS` | `1` |
| The ticket queue | `main.py` → `SAMPLES_PATH` | `data/sample/tickets.json` |

**Your own tickets:** edit `data/sample/tickets.json` — each ticket needs a `customer_id`, a `ticket` and an `expected` category.

### Usage

Run from the repo root:

```
python main.py
```

To see the parallel lookups save time (no API key needed):

```
python -m scripts.time_lookups
```

### Running the Tests

Run from the repo root:

```
# unit tests — no API key
python -m tests.unit.test_config
python -m tests.unit.test_schemas
python -m tests.unit.test_lookups
python -m tests.unit.test_fallback
python -m tests.unit.test_review

# integration tests — real API calls, needs OPENAI_API_KEY
python -m tests.integration.test_live_pipeline
```

`test_fallback.py` forces a real outage (a model pointed at a local port where nothing listens) and checks the fallback fires — and that a working chain and a real bug don't trigger it.

## Checklist Before You Call This Done

- [ ] The repo folder name matches the GitHub repo name exactly
- [ ] Code lives in `src/`, `main.py` at the repo root; runs with `python main.py`
- [ ] Every folder has one clear job, and you can say why it exists
- [ ] Only LCEL — no agent loop, no LangGraph — and you can say why none is needed
- [ ] The classifier returns a checked `TicketCategory`; every decision is logged with its reason
- [ ] `python -m scripts.time_lookups` shows the parallel step is really faster
- [ ] Only transient errors fall back; a bad key and a bug still fail loudly — all three proven by `test_fallback.py`
- [ ] One retry layer only (the client's), and you can explain why there's no `.with_retry()`
- [ ] Replies stream live, and the record is built from that same reply
- [ ] The unclear ticket and every fallback are flagged for a human, with the reason
- [ ] `tests/unit/` pass with no API key; `tests/integration/` pass with one
- [ ] `.env` is not in git
- [ ] You can explain every line of your own code, without help

## Publish It to GitHub

1. Copy README.md from this folder into your repo folder, and put your own name in its License line.
2. Check nothing secret is staged, then commit:

```
git status                    # .env and .venv/ must NOT appear
git add .
```

```
git commit -m "LangChainPro: an LCEL-only support-ticket pipeline"
```

3. On GitHub, create a new repo named exactly `LangChainPro-Branching-Parallel-And-Retry-Patterns-For-A-Real-Customer-Support-Pipeline` (no README — you have one). In the repo's **Description** box, paste the **Short description** line from the top of README.md. Then:

```
git branch -M main
git remote add origin <your-repo-url>
git push -u origin main
```

## Status

Not started. Track your progress in ../PROGRESS.md.

---

Stuck? Ask for **Hint 1** or **Hint 2** about the exact part you're stuck on. Only ask for the full code if you say **"Show me the solution."**